

Name: [REDACTED] Dorton

Instructor: Samuel Chukwuemeka

Objective: To convert a measurement from one unit to another unit

Measurement: Volume

First Given unit: Customary unit (Fluid ounce)

To Convert to: Metric unit (Milliliters)

Second Unit: Metric unit (Milliliters)

To Convert to: Customary unit (Fluid ounce)

Container used: Cantu Shea Butter Sulfate-Free Cleansing Cream Shampoo



First Conversion: **Convert 13.5 fluid ounce to milliliters**

From Given Tables

$$1 \text{ mL} = 10^{-3} \text{ L} = 0.001 \text{ L}$$

$$1 \text{ L} = 0.26417205 \text{ gal}$$

$$1 \text{ gal} = 128 \text{ fl oz}$$

Unity Fraction Method

$$13.5 \text{ fl oz} * \frac{\dots \text{gal}}{\dots \text{fl oz}} * \frac{\dots \text{L}}{\dots \text{gal}} * \frac{\dots \text{mL}}{\dots \text{L}}$$

$$13.5 \text{ fl oz} * \frac{1 \text{ gal}}{128 \text{ fl oz}} * \frac{1 \text{ L}}{0.26417205 \text{ gal}} * \frac{1 \text{ mL}}{0.001 \text{ L}}$$

$$\frac{13.5 \text{ mL}}{.0338140224}$$

$$399.2426527 \text{ mL}$$

Exact Value: 399.2426527 mL

Value on the container: 400 mL

The exact value was rounded up to the nearest whole number to get the value on the container.

Second Conversion: **Convert 400 milliliters to fluid ounce**

From Given Tables

$$1 \text{ mL} = 10^{-3} \text{ L} = 0.001 \text{ L}$$

$$1 \text{ L} = 0.26417205 \text{ gal}$$

$$1 \text{ gal} = 128 \text{ fl oz}$$

Unity Fraction Method

$$400 \text{ mL} * \frac{\dots \text{L}}{\dots \text{mL}} * \frac{\dots \text{gal}}{\dots \text{L}} * \frac{\dots \text{fl oz}}{\dots \text{gal}}$$

$$400 \text{ mL} * \frac{.001 \text{ L}}{1 \text{ mL}} * \frac{.26417205}{1 \text{ L}} * \frac{128}{1 \text{ gal}}$$

$$\frac{13.52560896 \text{ fl oz}}{1}$$

Exact Value: 13.52560896 fl oz

Value on the Container: 13.5 fl oz

The Exact Value was rounded to the nearest tenth to get the value on the container.